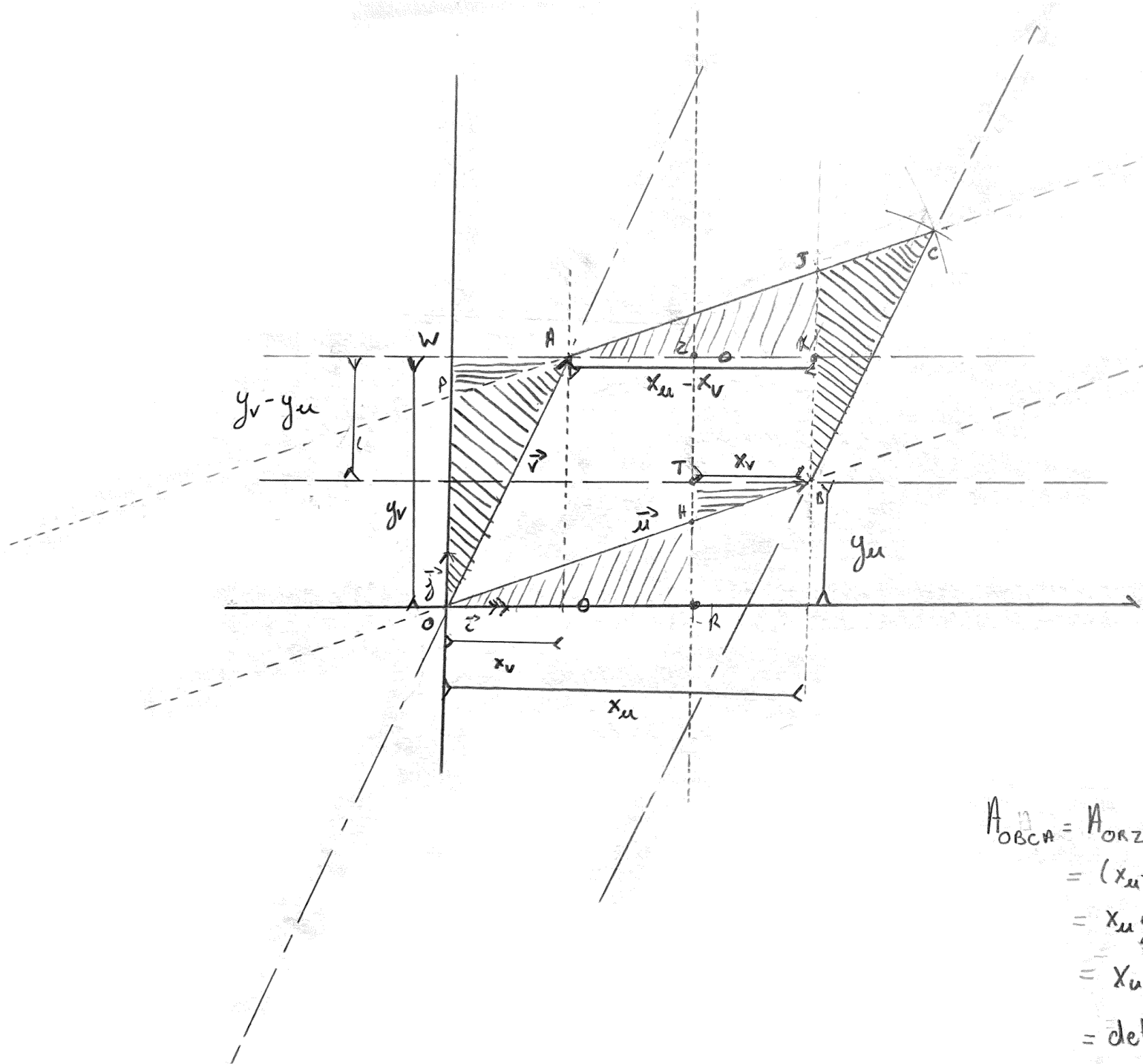




$$\begin{aligned}
 A_{\triangle ABC} &= A_{\text{ORZW}} + A_{\text{TBKZ}} \\
 &= (x_u - x_v)y_v + x_v(y_v - y_u) \\
 &= x_u y_v - x_v y_v + x_v y_v - x_v y_u \\
 &= x_u y_v - x_v y_u \\
 &= \det(\vec{u}, \vec{v})
 \end{aligned}$$